

Sparse-Graph Representations for Scalable Grouped Fused Regression

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Abstract

Grouped regression models estimate related but subgroup-specific linear parameters by combining sparsity with fusion penalties across a graph of groups. Although the statistical graph is often sparse, standard implementations can still materialize full pairwise fusion operators or augmented designs, making the computational representation much denser than the model. This paper studies sparse-graph representations for grouped fused regression and shows that the graph edge set can be preserved directly in the optimizer. For L1 fusion, an active-edge operator evaluates the same smoothed proximal-gradient fusion term using only nonzero graph edges. For L2 fusion, an active-edge augmented-design construction emits sparse triplets only for the modeled edges before solving the regularized path. These constructions preserve the legacy objective in the exact L1 and L2 cases while reducing the dominant fusion representation from all-pairs graph construction to edge-linear construction. The empirical comparisons intentionally hold the optimizer family fixed relative to the legacy grouped fused-regression solvers, so that the measured differences isolate graph representation rather than changes in optimization architecture. A chain-specialized L1 variant is also analyzed as a deliberate approximation that trades graph fidelity for additional efficiency. Across public grouped-regression benchmarks, the exact L1 operator matches legacy predictive accuracy while reducing median runtime by 2.26x and median peak memory by 10.7%. The chain variant gives a median 3.26x runtime reduction with an accuracy tradeoff. Across 19 L2 benchmark datasets, active-edge construction has no systematic loss in test error and reduces median fit time by 14.9x. The algorithms are implemented in the open-source R package `fuserplus`, with benchmark scripts and summaries provided for reproducibility.

Keywords: fused lasso, grouped regression, sparse graphs, structured sparsity, first-order optimization

1 Introduction

Many regression problems contain a natural grouping variable: patients may belong to disease subtypes, schools to districts, countries to regions, or time series to meters. Fitting a separate model in each group ignores shared signal, whereas fitting one global model ignores heterogeneity. Grouped fused regression addresses this middle ground by estimating one coefficient vector for each group and penalizing differences between coefficient vectors along a graph of group relationships.

Let $X_g \in \mathbb{R}^{n_g \times p}$ and $y_g \in \mathbb{R}^{n_g}$ denote the data for group g , and let $\beta_g \in \mathbb{R}^p$ be that group's linear coefficient vector. For a graph $G = (V, E)$ over k groups with nonnegative

edge weights w_{gh} , the estimators considered in this paper have the form

$$\min_{\beta_1, \dots, \beta_k} \sum_{g=1}^k \frac{1}{2n_g} \|y_g - X_g \beta_g\|_2^2 + \lambda \sum_{g=1}^k \|\beta_g\|_1 + \gamma \sum_{(g,h) \in E} w_{gh} \Omega(\beta_g - \beta_h). \quad (1)$$

For the L1 fused model, $\Omega(u) = \|u\|_1$; for the L2 fused model, $\Omega(u) = \frac{1}{2} \|u\|_2^2$ after the standard augmented-design reformulation. Optional group-specific intercepts are included in the software implementation but are excluded from sparsity and fusion penalties. This formulation is a direct descendant of the lasso, fused lasso, grouped and elastic-net regularization, graph-structured fused optimization, and subgroup regression models reviewed below. The L1 solver uses the standard Nesterov smoothing strategy for nonsmooth penalties (Nesterov, 2005).

The central computational issue is a mismatch between the statistical object and the solver representation. A complete graph has $k(k-1)/2$ edges, but practical graphs are frequently chains, trees, nearest-neighbor graphs, or domain-informed sparse networks. If the objective is defined on only m active edges, a solver that builds all pairwise fusion constraints and then uses zero weights for inactive edges changes the computational scaling of the problem without changing its statistical content. This matters as k grows: the modeling assumption is sparse, but the legacy representation is still quadratic.

The paper therefore focuses on a controlled algorithmic comparison rather than on replacing the optimizer family. For L1 fusion, both the legacy solver and the exact active-edge solver use the same smoothed proximal-gradient framework. For L2 fusion, both the legacy solver and the active-edge solver use the same augmented-design reduction before fitting the regularized path. Alternative methods such as ADMM and split-Bregman solvers are important for nonsmooth fused penalties, but they change the optimization architecture and introduce additional tuning and implementation choices. The comparisons here are designed to isolate the effect of preserving the sparse graph inside the existing grouped fused-regression optimization framework.

The contribution of this paper is an algorithmic representation of grouped fused regression that makes the sparse graph a first-class object in the optimizer. The key idea is to represent the fused-regression objective in a way that respects the graph on which the penalty is defined. First, the paper gives an exact active-edge L1 operator form that evaluates the smoothed proximal-gradient fusion term only over active edges. Second, it gives an exact active-edge L2 augmented-design construction that emits sparse triplets only for nonzero graph edges before delegating the regularized regression path to `glmnet` (Friedman et al., 2010). Third, it separates these exact reformulations from approximation-oriented solver modes, including a chain-specialized L1 approximation that collapses a graph to $k-1$ chain edges using a depth-first traversal, motivated by chain-structured fused lasso solvers. The result is a reusable design pattern for multi-task fused regression and other graph-penalized models whose penalties decompose over graph edges.

The main contributions are:

- a sparse-graph representation principle for grouped fused regression solvers;
- exact active-edge L1 and L2 reformulations that preserve the legacy objective while avoiding zero-weight all-pair construction;

- a clear separation between exact sparse-graph solvers and approximate high-efficiency chain variants;
- a controlled empirical design that keeps the optimizer family fixed and isolates the effect of graph representation;
- empirical comparisons showing when fused regression improves over unfused `glmnet` baselines and when active-edge construction improves over legacy fused solvers; and
- a reproducible open-source implementation with benchmark scripts, manifests, and figure-generation artifacts.

The algorithms are implemented in the R package `fuserplus`. The current benchmark suite uses public grouped regression datasets and synthetic scaling experiments to compare prediction error, wall time, and peak resident memory. The results show that graph-aware construction can deliver substantial runtime and memory reductions without changing the statistical objective for the exact operator and L2 variants.

2 Related Work

The fused lasso extends the lasso by combining sparsity with penalties on ordered coefficient differences (Tibshirani, 1996; Tibshirani et al., 2005). This places grouped fused regression in a broader family of structured regularizers, including the elastic net, group lasso, generalized lasso, and graph trend filtering (Zou and Hastie, 2005; Yuan and Lin, 2006; Tibshirani and Taylor, 2011; Wang et al., 2016). Specialized algorithms for one-dimensional signal approximation, including path algorithms for the fused lasso signal approximator, exploit the ordered chain structure of that problem (Hoeftling, 2010). Split Bregman and ADMM methods provide another important optimization line for nonsmooth fused penalties (Ye and Xie, 2011; Boyd et al., 2011). General graph fusion is more flexible but computationally harder because the penalty couples many coefficient blocks.

Graph-structured multi-task regression extended these ideas to arbitrary task graphs and developed smoothed proximal-gradient methods for the general fused lasso (Chen et al., 2010, 2012a). Related multi-task and scientific applications use graph-guided fusion for eQTL mapping, additive modeling, gene regulatory network inference, and subgroup-specific regression (Chen et al., 2012b; Petersen et al., 2016; Omranian et al., 2016). Network lasso methods (Hallac et al., 2015), graph trend filtering, DFS fused lasso denoising (Tansey et al., 2018), and k-nearest-neighbor fused lasso estimators (Padilla et al., 2020) further emphasize that the graph itself is part of the modeling problem: it may come from prior knowledge, distances between groups, or other data-driven structure.

The present work focuses on how grouped fused regression objectives should be represented and solved when the graph is sparse. This is more than an implementation detail: the graph is part of the modeling assumption, and a dense all-pairs representation discards that assumption at the computational layer. The active-edge constructions developed below restore the intended edge-level representation while preserving the legacy objective in the exact L1 and L2 cases.

Table 1 positions the paper relative to the main modeling and optimization ideas in the literature, rather than comparing software packages. Prior work supplies the sparsity and

Table 1: Literature positioning for sparse-graph grouped fused regression.

Literature line	Connection to this paper
Lasso and fused lasso (Tibshirani, 1996; Tibshirani et al., 2005)	Introduce the core sparsity and fusion penalties used in graph-fused regression. This paper studies how sparse fusion graphs can be represented efficiently in grouped regression.
Generalized lasso and graph trend filtering (Tibshirani and Taylor, 2011; Wang et al., 2016)	Provide broad operator and graph-difference frameworks for structured variation. The focus here is the active-edge construction needed by paired L1/L2 grouped fused-regression solvers.
Chain and DFS fused-lasso algorithms (Hoeffing, 2010; Tansey et al., 2018)	Use special graph structure to obtain fast one-dimensional or tree/DFS-based fused estimates. This paper treats chain reduction as an approximation, while the active-edge solver keeps the original sparse graph.
Split-Bregman and ADMM approaches (Ye and Xie, 2011; Boyd et al., 2011)	Use auxiliary variables for nonsmooth penalties and solve alternating subproblems. The comparisons in this paper keep the smoothed first-order optimization framework fixed to isolate the effect of graph representation.
Graph-guided multitask regression (Chen et al., 2010, 2012a,b)	Uses task graphs and smoothed proximal-gradient methods for fused multitask penalties. This paper applies the same broad optimization setting to grouped regression with active-edge representations for L1 and L2 fusion.
Network lasso and graph-based fusion (Hallac et al., 2015)	Penalizes graph-edge differences to cluster or share information over networks. This supports the same edge-based modeling principle, but the objective and optimizer architecture differ from the controlled comparison used here.
K-nearest-neighbor fused lasso (Padilla et al., 2020)	Builds data-driven nearest-neighbor graphs for adaptive nonparametric fused regression. This motivates sparse graph construction; the paper studies how such graphs are represented inside grouped fused-regression solvers.
Joint lasso for subgroup regression (Dondelinger et al., 2020)	Estimates grouped regressions with sparsity and fusion across groups. This paper preserves that objective while replacing all-pair graph construction with active-edge construction.

fusion penalties, general graph-difference formulations, data-driven graph construction, and smoothed first-order optimization. The present paper keeps the grouped fused-regression objective and optimizer family fixed, and studies sparse edge representations for L1 and L2 fusion.

Figures 1 and 2 summarize the main computational distinction between the legacy fused-regression representation and the sparse-graph representation used in `fuserplus`. In both cases the statistical model is defined by a graph on the k groups, with an edge weight indicating how strongly two subgroup coefficient vectors should be encouraged to agree. The legacy construction first expands this modeling graph into a dense algebraic object over many group pairs. That expansion is benign for small k , but it becomes the dominant cost when the intended graph is sparse, because computation is then driven by the number of possible group pairs rather than by the number of modeled edges.

Figure 1 describes the L1 fusion case. Panel A begins with the intended fusion graph: nodes are regression groups and each drawn edge marks a coefficient difference penalized by the model. The legacy full-edge construction expands that graph into an all-pairs layout for the combined L1 constraint matrix; in the four-group sketch, it introduces pair slots that are absent from the modeled graph. The exact active-edge operator keeps the original graph and evaluates the nonsmooth fusion penalty only along its m nonzero edges. The chain approximation makes a different tradeoff by replacing the original edge set with a chain before fitting. Panel B then shows the computational consequence: the active-edge L1 representation avoids the all-pairs constraint matrix, so fusion-side scaling depends on the graph edges rather than powers of k .

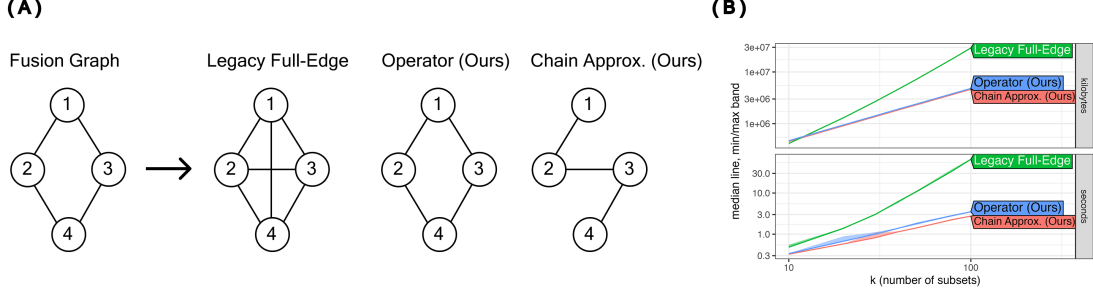


Figure 1: Conceptual overview of L1 sparse-graph fusion. A: fusion graph construction. The legacy full-edge construction associated with Dondelinger et al. (2020) forms a combined constraint matrix over all group pairs, while the active-edge operator works directly on the sparse graph. The chain approximation replaces the modeled edge set with a chain before fitting. B: asymptotic complexity. The synthetic example fixes $p = 120$ and $n_{\text{group,train}} = 35$, so at $k = 100$ the total training sample size is $n = 3500$. With a sparse chain graph and fixed remaining hyperparameters, the active-edge representation changes the dominant fusion cost from $O(pk^3)$ to $O(pm)$; the chain approximation is plotted alongside the exact operator as the chain-based L1 variant.

Figure 2 gives the analogous picture for L2 fusion. In the quadratic case, fusion can be represented by augmenting the regression design with pseudo-observations that penalize differences between subgroup coefficients. The legacy construction emits augmented rows for the expanded set of group pairs; the active-edge construction emits sparse triplets only for edges with nonzero fusion weight. Thus the same modeling graph produces a much smaller linear-algebra problem whenever $m \ll k^2$. These two figures motivate the solver descriptions in Section 4 and the scaling experiments reported below: the central question is whether preserving the edge-level graph representation yields the same fitted objective at lower runtime and memory cost.

3 Model and Legacy Computation

The L1 solver in `fuserplus` follows a smoothed proximal-gradient strategy for Equation 1. Let $B = [\beta_1, \dots, \beta_k]$ be the $p \times k$ coefficient matrix. The legacy implementation builds a constraint matrix with columns for sparsity and fusion constraints, then uses a smoothed dual approximation to compute the gradient contribution of the nonsmooth penalty. The update has the generic accelerated form

$$A^* = \Pi_{[-1,1]} \left(\frac{BC}{\mu} \right), \quad (2)$$

$$\nabla f_\mu(B) = \nabla \ell(B) + A^* C^\top, \quad (3)$$

$$B^{t+1} = W^t - L^{-1} \nabla f_\mu(B^t), \quad (4)$$

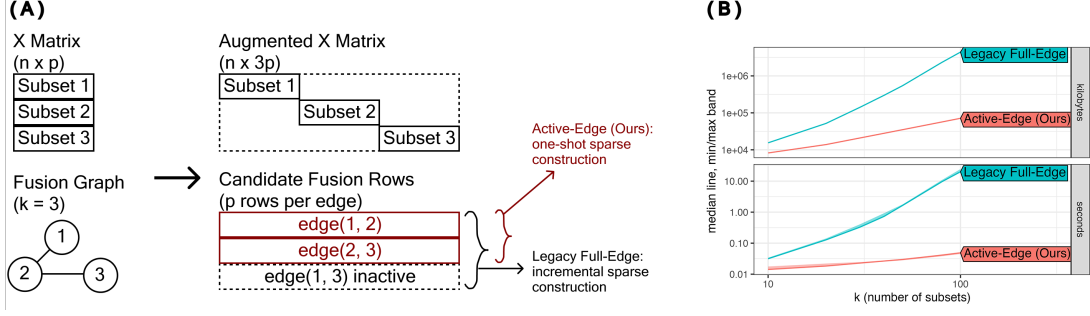


Figure 2: Conceptual overview of L2 sparse-graph fusion. A: augmented matrix construction. The legacy full-edge construction associated with Dondelinger et al. (2020) forms augmented rows for all group pairs, while the active-edge builder emits only sparse triplets for graph edges with nonzero weight. B: asymptotic complexity. The synthetic example fixes $p = 120$ and $n_{\text{group,train}} = 35$, so at $k = 100$ the total training sample size is $n = 3500$. With a sparse chain graph and fixed remaining hyperparameters, active-edge construction changes the dominant fusion-block cost from $O(pk^2)$ to $O(pm)$.

where C encodes the sparsity and fusion operators, μ is the smoothing parameter, L is a Lipschitz bound, and W^t is the accelerated iterate. When C contains all group pairs, storage and multiplication include $O(pk^2)$ fusion terms for a complete representation.

More explicitly, the L1 implementation uses the dual identity $|z| = \max_{|a| \leq 1} az$ and replaces the nonsmooth absolute-value terms by clipped smoothed duals. For sparsity and active fusion edges this gives

$$A_s^* = \text{clip} \left(\frac{\lambda B_s}{\mu}, -1, 1 \right), \quad A_{uv}^* = \text{clip} \left(\frac{\gamma w_{uv}(\beta_u - \beta_v)}{\mu}, -1, 1 \right),$$

where B_s excludes intercept rows. The code uses the step-size upper bound

$$L_U = \lambda_{\max} + \frac{\lambda^2 + 2\gamma^2 d_{\max}^{(2)}}{\mu},$$

with $\lambda_{\max} = \sum_g \lambda_{\max}(X_g^\top X_g / n_g)$ under scaled group losses and $d_{\max}^{(2)}$ any upper bound on $\max_g \sum_{h:(g,h) \in E} w_{gh}^2$ for the active fusion graph. This quantity is bounded by the usual weighted degree when weights are normalized to lie in $[0, 1]$.

3.1 Optimization Guarantee for Smoothed L1

The guarantees below are deterministic statements for a fixed graph, fixed hyperparameters, and fixed train/validation split. They concern objective representation and optimization, not statistical recovery under a data generating model. The first guarantee specializes the standard consequences of Nesterov smoothing and accelerated proximal-gradient convergence to the grouped fused-regression notation used here (Nesterov, 2005; Beck and Teboulle, 2009).

Proposition 1 (Lipschitz and convergence guarantee for smoothed L1)

Let $F_\mu(B)$ be the Nesterov-smoothed L1 grouped fused objective, before final postprocessing. Let $B_\mu^\star$ minimize F_μ . If the APG update uses any L_U satisfying

$$L_U \geq \lambda_{\max} + \frac{\lambda^2 + 2\gamma^2 d_{\max}^{(2)}}{\mu},$$

then ∇F_μ is L_U -Lipschitz and the accelerated iterates satisfy

$$F_\mu(B^{(t)}) - F_\mu(B_\mu^\star) \leq \frac{2L_U \|B^{(0)} - B_\mu^\star\|_F^2}{(t+1)^2}.$$

Moreover, if N_c is the number of scalar sparsity and fusion constraints in the active-edge representation, then the nonsmoothed objective F and its smoothed version obey

$$0 \leq F(B) - F_\mu(B) \leq \frac{\mu N_c}{2}$$

for every coefficient matrix B .

Proof The least-squares part has Lipschitz constant bounded by $\lambda_{\max}$ under the scaled group losses. The smoothed sparsity operator contributes at most λ^2/μ to the gradient Lipschitz constant. For the fusion operator, each edge difference has two incident group blocks; the squared operator norm is bounded by $2d_{\max}^{(2)}$, giving the fusion contribution $2\gamma^2 d_{\max}^{(2)}/\mu$. Summing these bounds gives L_U . The stated $O(t^{-2})$ rate is the standard accelerated-gradient guarantee for convex differentiable objectives with L_U -Lipschitz gradient. Finally, each scalar absolute-value term satisfies $0 \leq |z| - \max_{|a| \leq 1} \{az - \mu a^2/2\} \leq \mu/2$; summing over the N_c scalar constraints gives the smoothing gap. ■

Proposition 2 (Equivalence of full-edge and active-edge L1 updates)

Fix a weighted graph G and let E denote the edges with positive weight. For smoothing parameter $\mu > 0$, suppose the dense smoothed proximal-gradient implementation includes one fusion column for every pair but assigns zero weight to inactive pairs. The active-edge operator implementation computes exactly the same smoothed fusion gradient as the dense implementation after removing the zero-weight columns.

Proof Let e_u be the u th coordinate vector in $\mathbb{R}^k$. For any pair $1 \leq u < v \leq k$, define the smoothed dual block

$$P_{uv}(B) = \text{clip} \left(\frac{\gamma w_{uv}(\beta_u - \beta_v)}{\mu}, -1, 1 \right).$$

The dense fusion contribution to the smoothed gradient can be written as

$$\Delta_{\text{dense}}(B) = \sum_{1 \leq u < v \leq k} \gamma w_{uv} P_{uv}(B) (e_u - e_v)^\top.$$

Algorithm 1 Legacy Full-Edge Smoothed APG (old.11)

Require: grouped data, $G, \lambda, \gamma, \mu, T, \varepsilon$

- 1: Construct full fusion matrix $C_{\text{fusion}} \in \mathbb{R}^{k \times \binom{k}{2}}$
- 2: Construct $C = [\lambda I_k, \gamma C_{\text{fusion}}]$
- 3: Initialize $B^{(0)} = 0, W^{(0)} = 0, S^{(0)} = 0$
- 4: **for** $t = 1, \dots, T$ **do**
- 5: Compute $\nabla \ell(B^{(t-1)})$ groupwise
- 6: $A^* \leftarrow \text{clip}([B_s^{(t-1)} C_s, B_f^{(t-1)} C_f] / \mu, -1, 1)$
- 7: $\nabla r_\mu(B^{(t-1)}) \leftarrow A^* C^\top$
- 8: $H_t \leftarrow \nabla \ell(B^{(t-1)}) + \nabla r_\mu(B^{(t-1)})$
- 9: Compute L_U from the dense max-degree proxy
- 10: Update $\tilde{B}^{(t)}, S^{(t)}, W^{(t)}$
- 11: **if** $\|\tilde{B}^{(t)} - B^{(t-1)}\|_1 < \varepsilon p$ **then**
- 12: **break**
- 13: **end if**
- 14: $B^{(t)} \leftarrow \tilde{B}^{(t)}$
- 15: **end for**
- 16: Soft-threshold non-intercept rows by λ/n_g
- 17: **return** $\hat{B}$

If $w_{uv} = 0$, then the corresponding summand is zero. Therefore

$$\Delta_{\text{dense}}(B) = \sum_{(u,v) \in E} \gamma w_{uv} P_{uv}(B) (e_u - e_v)^\top = \Delta_{\text{active}}(B).$$

Algorithm 2 computes this last expression by adding $\gamma w_{uv} P_{uv}(B)$ to column u and subtracting it from column v for each active edge. Hence the dense and active-edge implementations produce the same smoothed fusion-gradient contribution, and therefore the same L1 update when the remaining APG quantities are held fixed. $\blacksquare$

3.2 Full-Edge L2 Augmented Design

The L2 solver uses a different reduction. It constructs an augmented response and design matrix,

$$\tilde{y} = \begin{bmatrix} y \\ 0 \end{bmatrix}, \quad \tilde{X} = \begin{bmatrix} X_{\text{block}} \\ X_{\text{fusion}} \end{bmatrix},$$

where X_{block} is block diagonal over groups and X_{fusion} contains, for each group pair, rows proportional to

$$\sqrt{\gamma w_{gh}} (\beta_g - \beta_h).$$

Let p_{eff} denote the number of coefficient coordinates included in this L2 fusion block after applying the intercept convention. The resulting lasso problem is passed to `glmnet`. A full all-pairs construction creates $O(p_{\text{eff}} k^2)$ fusion rows even when only $m = |E|$ edges are active.

Proposition 3 (Equivalence of active-edge L2 augmented rows) *For a fixed graph G , penalty γ , and scaling convention, the active-edge L2 augmented design is equivalent to the legacy all-pairs augmented design after deleting all zero-weight pair rows and applying a row permutation.*

Proof Each nonzero edge (u, v) contributes p_{eff} augmented rows with $+\sqrt{\gamma w_{uv}}$ in coefficient block u and $-\sqrt{\gamma w_{uv}}$ in coefficient block v , with zero augmented response. These rows contribute

$$\gamma w_{uv} \|\beta_u - \beta_v\|_2^2$$

to the augmented least-squares objective. Therefore the all-pairs fusion contribution satisfies

$$\sum_{1 \leq u < v \leq k} \gamma w_{uv} \|\beta_u - \beta_v\|_2^2 = \sum_{(u,v) \in E} \gamma w_{uv} \|\beta_u - \beta_v\|_2^2,$$

because inactive pairs have $w_{uv} = 0$. The active-edge builder constructs exactly the rows corresponding to the right-hand side using sparse triplets. Permuting those rows changes only their order, not the sum of squared residuals, so the objective minimized by `glmnet` is unchanged. $\blacksquare$

Proposition 4 (Sparse-graph construction cost)

Let k be the number of groups, p the number of penalized coefficients per group in the L1 fusion operator, $m = |E|$ the number of nonzero fusion edges, and $q = \binom{k}{2}$ the number of all possible group pairs. Let p_{eff} be the number of coefficients penalized by the L2 fusion block after the intercept convention is applied. In one L1 smoothed-gradient evaluation, a dense all-pairs operator performs fusion work proportional to pq after construction, whereas the active-edge operator performs fusion work proportional to pm . In the L2 augmented-design construction, the fusion block decreases from $p_{\text{eff}}q$ augmented rows to $p_{\text{eff}}m$ augmented rows.

Proof For L1 fusion, each represented edge requires a p -dimensional coefficient difference and a p -dimensional scatter back to the incident groups. Thus the fusion-side work scales as

$$W_{\text{L1,full}} \asymp pq = p \binom{k}{2}, \quad W_{\text{L1,active}} \asymp pm.$$

The dense all-pairs representation iterates over q candidate pairs, including zero-weight inactive pairs, while the active-edge representation iterates only over the m nonzero graph edges.

For L2 fusion, each represented edge contributes p_{eff} augmented rows, one for each penalized coefficient coordinate. Therefore the fusion block sizes are

$$R_{\text{L2,full}} = p_{\text{eff}}q = p_{\text{eff}} \binom{k}{2}, \quad R_{\text{L2,active}} = p_{\text{eff}}m.$$

Deleting zero-weight pairs gives the stated row reduction, with p_{eff} determined by the intercept convention. $\blacksquare$

4 Sparse-Graph Solvers

4.1 Exact L1 Operator Solver

Here $B = [\beta_1, \dots, \beta_k]$ is the grouped coefficient matrix, with one column for each group. The exact L1 operator solver replaces the dense fusion matrix representation with edge lists $(u_e, v_e, w_e)_{e=1}^m$, where $m = |E|$ is the number of active fusion edges, u_e and v_e are the incident groups for edge e , and $w_e > 0$ is its fusion weight. At each iteration, the solver computes the fusion differences only on active edges,

$$D_e = \beta_{u_e} - \beta_{v_e}, \quad P_e = \text{clip} \left(\frac{\gamma w_e D_e}{\mu}, -1, 1 \right),$$

where D_e is the coefficient difference vector and P_e is its clipped smoothed dual block. The solver scatters $\gamma w_e P_e$ back to the incident group columns with opposite signs. The sparsity part of the update remains unchanged. This representation has the same objective and first-order updates as the active-edge version of the legacy smoothed proximal-gradient solver, but the fusion-gradient work scales with pm rather than pk^2 .

The implementation also supports block processing of edges. This is a practical memory safeguard for large p or m , since intermediate edge-difference matrices can otherwise dominate the working set. The operator solver is therefore intended as the general exact path for sparse graphs.

Active-edge gradient assembly. The operator implementation initializes $\Delta_f = 0$ and loops over active edges $e = 1, \dots, m$. For each edge, it computes $d_e = \beta_{u_e} - \beta_{v_e}$ and $p_e = \text{clip}((\gamma w_e / \mu) d_e, -1, 1)$, then performs the endpoint updates

$$\Delta_f(:, u_e) \leftarrow \Delta_f(:, u_e) + \gamma w_e p_e, \quad \Delta_f(:, v_e) \leftarrow \Delta_f(:, v_e) - \gamma w_e p_e.$$

The accumulated matrix Δ_f is the fusion contribution added to the likelihood and sparsity gradients in the accelerated update.

Optimization algorithm. The implementation used in the experiments is the following active-edge APG procedure. Here T is the maximum number of APG iterations and ε is the stopping tolerance. It is the central exact L1 routine in **fuserplus**.

Algorithm 2 Active-Edge Operator APG (operator)

Require: grouped data, edge list E , $\lambda, \gamma, \mu, T, \varepsilon$
 1: Initialize $B^{(0)} = 0$, $W^{(0)} = 0$, $S^{(0)} = 0$
 2: **for** $t = 1, \dots, T$ **do**
 3: Compute groupwise likelihood gradient $\nabla \ell(B^{(t-1)})$
 4: $A_s^* \leftarrow \text{clip}(\lambda B_s^{(t-1)} / \mu, -1, 1)$
 5: Initialize fusion accumulator $\Delta_f \leftarrow 0$
 6: **for** each edge $e = (u_e, v_e, w_e) \in E$ **do**
 7: $d_e \leftarrow \beta_{u_e}^{(t-1)} - \beta_{v_e}^{(t-1)}$
 8: $p_e \leftarrow \text{clip}((\gamma w_e / \mu) d_e, -1, 1)$
 9: $\Delta_f(:, u_e) \leftarrow \Delta_f(:, u_e) + \gamma w_e p_e$
 10: $\Delta_f(:, v_e) \leftarrow \Delta_f(:, v_e) - \gamma w_e p_e$
 11: **end for**
 12: $H_t \leftarrow \nabla \ell(B^{(t-1)}) + \lambda A_s^* + \Delta_f$
 13: Compute L_U on active graph; update $\tilde{B}^{(t)}, S^{(t)}, W^{(t)}$
 14: **if** $\|\tilde{B}^{(t)} - B^{(t-1)}\|_1 < \varepsilon p$ **then**
 15: **break**
 16: **end if**
 17: $B^{(t)} \leftarrow \tilde{B}^{(t)}$
 18: **end for**
 19: Soft-threshold non-intercept rows by λ/n_g
 20: **return** $\hat{B}$

4.2 Chain-Specialized L1 Approximation

The chain-specialized solver is a deliberately approximate method. It builds a depth-first traversal of a maximum spanning tree or of the thresholded graph, reorders the groups along that traversal, and keeps only the $k - 1$ adjacent chain edges. Let π denote the resulting group ordering and let $w = (w_1, \dots, w_{k-1})$ denote the adjacent chain weights after reordering. The fusion operator can then be computed with vectorized adjacent differences:

$$D_j = \beta_j - \beta_{j+1}, \quad j = 1, \dots, k - 1.$$

The vectorized implementation forms

$$D = B_{:,1:(k-1)} - B_{:,2:k}, \quad P = \text{clip}((\gamma/\mu)D \text{diag}(w), -1, 1),$$

and accumulates $\gamma P \text{diag}(w)$ on neighboring chain endpoints. This reduces the edge budget to the minimum needed to connect the groups. It does not optimize the same objective as the original graph unless the graph is already the selected chain. It is useful when the main objective is maximum speed and memory reduction and the user is willing to accept a small loss in graph fidelity.

Proposition 5 (Chain approximation target)

Let $G_\pi = (V, E_\pi)$ be the chain graph produced by the DFS/MST ordering, with edge weights inherited from the original graph. The chain-specialized solver applies the exact active-edge

L1 update to the smoothed fused objective on G_π . Define the original graph-fusion penalty as

$$R_G(B) = \gamma \sum_{(u,v) \in E} w_{uv} \|\beta_u - \beta_v\|_1,$$

and define $R_{G_\pi}(B)$ analogously on the chain. Then for any coefficient matrix B and any chain with $E_\pi \subseteq E$,

$$0 \leq R_G(B) - R_{G_\pi}(B) = \gamma \sum_{(u,v) \in E \setminus E_\pi} w_{uv} \|\beta_u - \beta_v\|_1.$$

Thus the approximation error is exactly the omitted-edge fusion penalty for the fitted coefficient matrix.

Proof After the groups are reordered by π , the vectorized adjacent-difference operator is the active-edge operator for the chain edges (π_j, π_{j+1}) , $j = 1, \dots, k-1$. The scatter updates therefore match the L1 active-edge update on G_π . If $E_\pi \subseteq E$, subtracting the chain penalty from the original graph penalty leaves precisely the sum over omitted edges, which is nonnegative. $\blacksquare$

Optimization algorithm. The chain-specialized code first builds the DFS/MST order and adjacent chain weights, then runs a separate vectorized APG routine rather than calling the general edge loop. In Algorithm 3, $\eta = 1/L_U$ is the step size, H_t is the smoothed gradient at iteration t , $R^{(t)}$ is the accumulated scaled-gradient term, $Z^{(t)} = -R^{(t)}$ is the aggregate point used in the acceleration step, and $W^{(t)}$ is the accelerated iterate:

4.3 Active-Edge L2 Augmented Design

The L2 active-edge solver preserves the augmented-design strategy but changes how the design is built. Instead of allocating all pairwise fusion blocks and then assigning zero-weight rows, it finds the upper-triangular nonzero entries of G and builds sparse triplets only for those edges. As above, p_{eff} denotes the number of coefficient coordinates included in the L2 fusion block, and $n_{\text{fusion}} = p_{\text{eff}}m$ is the number of active fusion rows. The active-edge construction keeps the same augmented-design scaling: $\sqrt{\gamma(n + n_{\text{fusion}})}\sqrt{w_{gh}}$ multiplies the positive and negative entries in the fusion rows. The final augmented design is passed to **glmnet** (Friedman et al., 2010) with the same lambda correction factor as the original wrapper.

This change leaves the fitted objective unchanged for a given active graph but reduces the number of constructed fusion rows from $p_{\text{eff}}k(k-1)/2$ to $p_{\text{eff}}m$. The largest expected gains occur when $m \ll k(k-1)/2$.

Optimization algorithm. The L2 path uses **glmnet** after building an augmented sparse regression problem. The fitted vector θ stacks the groupwise coefficients and is reshaped into the coefficient matrix B after fitting. Let p_B denote the number of coefficient rows in B after applying the intercept convention:

Table 2 summarizes the asymptotic costs implied by the preceding solver descriptions. The purpose is to separate the common groupwise regression cost, represented by $O(np)$

Algorithm 3 Chain-Specialized APG (`chain_specialized`)

Require: grouped data, G , chain settings, $\lambda, \gamma, \mu, T, \varepsilon$

- 1: Build DFS/MST chain order π and adjacent chain weights $(w_1, \dots, w_{k-1})$
 - 2: Reorder groups by π ; initialize $B^{(0)} = 0$, $W^{(0)} = 0$, $R^{(0)} = 0$
 - 3: Compute chain Lipschitz bound L_U and set $\eta \leftarrow 1/L_U$
 - 4: **for** $t = 1, \dots, T$ **do**
 - 5: Compute groupwise likelihood gradient $\nabla \ell(B^{(t-1)})$
 - 6: $A_s^* \leftarrow \text{clip}(\lambda B_s^{(t-1)} / \mu, -1, 1)$
 - 7: $D \leftarrow B_{:,1:(k-1)}^{(t-1)} - B_{:,2:k}^{(t-1)}$
 - 8: $P \leftarrow \text{clip}\left(\frac{\gamma D \text{diag}(w)}{\mu}, -1, 1\right)$
 - 9: $C \leftarrow \gamma P \text{diag}(w)$
 - 10: Initialize fusion accumulator $\Delta_f \leftarrow 0$
 - 11: $\Delta_f(:, 1 : (k-1)) += C$, $\Delta_f(:, 2 : k) -= C$
 - 12: $H_t \leftarrow \nabla \ell(B^{(t-1)}) + \lambda A_s^* + \Delta_f$
 - 13: $\tilde{B}^{(t)} \leftarrow W^{(t-1)} - \eta H_t$
 - 14: $R^{(t)} \leftarrow R^{(t-1)} + \frac{t\eta}{2} H_t$
 - 15: $Z^{(t)} \leftarrow -R^{(t)}$
 - 16: $W^{(t)} \leftarrow (t\tilde{B}^{(t)} + 2Z^{(t)}) / (t+2)$
 - 17: **if** $\|\tilde{B}^{(t)} - B^{(t-1)}\|_1 < \varepsilon p$ **then**
 - 18: **break**
 - 19: **end if**
 - 20: $B^{(t)} \leftarrow \tilde{B}^{(t)}$
 - 21: **end for**
 - 22: Soft-threshold non-intercept rows by λ/n_g
 - 23: Reorder columns back to original group order
 - 24: **return** $\hat{B}$
-

terms, from the cost of the fusion representation. The L1 time entries describe one APG iteration, while the L2 time entries describe one pass over the augmented design. For L2, the table uses p_{eff} because the intercept convention determines how many coefficient coordinates are included in the fusion block. The active-edge L1 memory bound assumes the block processing described above, so edge-difference workspaces are not materialized for all m edges at once.

5 Experimental Design

5.1 Datasets

The empirical evaluation uses public grouped-regression datasets spanning different sample sizes, feature dimensions, and numbers of groups. Across the processed collection, the number of observations ranges from 200 to 220,908, the number of features from 3 to 100, and the number of groups from 10 to 3,058. The sources include UCI datasets (Chen, 2017; Redmond, 2002; Trindade, 2015), Rdatasets mirrors (Arel-Bundock, 2026), Our World in

Algorithm 4 L2 Augmented-Design Solver with `glmnet`

Require: grouped data (X, y, groups) , graph G , penalties λ, γ , intercept and loss-scaling conventions

- 1: Build block-diagonal design X_{block} over groups
- 2: Extract active weighted edges $E = \{(u, v, w_{uv}) : u < v, G_{uv} = w_{uv} > 0\}$
- 3: Let $\mathcal{P}$ be the coefficient coordinates included in L2 fusion, with $|\mathcal{P}| = p_{\text{eff}}$
- 4: Initialize sparse triplet lists for X_{fusion} and set row counter $a \leftarrow 0$
- 5: **for** each active edge $(u, v) \in E$ **do**
- 6: **for** each coefficient coordinate $r \in \mathcal{P}$ **do**
- 7: $a \leftarrow a + 1$
- 8: Add triplet $(a, \text{col}(u, r), +\sqrt{w_{uv}})$
- 9: Add triplet $(a, \text{col}(v, r), -\sqrt{w_{uv}})$
- 10: **end for**
- 11: **end for**
- 12: Set $n_{\text{fusion}} \leftarrow a$ and materialize sparse matrix X_{fusion} from the triplets
- 13: Scale fusion block by $\sqrt{\gamma(n + n_{\text{fusion}})}$
- 14: Form augmented system $\tilde{X} = [X_{\text{block}}; X_{\text{fusion}}]$, $\tilde{y} = [y; 0]$
- 15: Fit θ with `glmnet` on $(\tilde{X}, \tilde{y})$ using the corrected λ
- 16: Reshape θ to $B \in \mathbb{R}^{p_B \times k}$
- 17: **return** B

Table 2: Dominant representation costs of the solver variants. Here n is total sample size, p is feature count, p_{eff} is the number of coordinates included in the L2 fusion block, k is group count, $m = |E|$ is the number of active fusion edges, and $q = \binom{k}{2}$. The table isolates graph-representation costs and omits lower-order constants from R bookkeeping and solver overhead.

Method	Time	Memory
L1 full edge	$O(np + pkq)$	$O(np + pk + pq + kq)$
L1 active-edge operator	$O(np + pm)$	$O(np + pk + m)$
L1 chain-specialized	$O(np + pk)$	$O(np + pk)$
L2 full pairwise	$O(np + p_{\text{eff}}q)$	$O(np + p_{\text{eff}}q)$
L2 active-edge	$O(np + p_{\text{eff}}m)$	$O(np + p_{\text{eff}}m)$

Data CO2 data (Ritchie et al., 2023), World Bank indicators (World Bank, 2026), NYC TLC trip records (New York City Taxi and Limousine Commission, 2022), OpenML (OpenML, 2026), and additional public examples and package datasets (PyMC Development Team, 2026; Pinheiro and Bates, 2000; Raudenbush and Bryk, 2002).

The final L1 benchmark includes 12 datasets with five outer folds for each dataset. The final L2 benchmark includes 19 datasets with five outer folds for each dataset, including 11 datasets with at least 50 groups. The L1 and L2 panels differ because the two solver families have different computational budgets and failure modes on the largest grouped

problems. The complete union of datasets used in the reported L1 and L2 summaries is listed in Supplementary Section A.

5.2 Evaluation Protocol

All reported real-data results use grouped outer cross-validation with five stratified folds and random seed 20260221 (Kohavi, 1995; Varma and Simon, 2006). Splits are made at the observation level while preserving group representation where possible; datasets with very small groups are filtered using a minimum group size of seven. All graph construction, hyperparameter selection, and standardization are performed inside the outer-training fold so that the held-out test fold is not used in model selection.

For each outer fold, a baseline sparsity penalty is selected with fusion disabled, $\gamma = 0$, on the outer-training data by `cv.glmnet` with three inner folds and `lambda.min`. We denote this selected sparsity penalty by λ in Equation 1. Holding this λ fixed, a separate inner train-validation split of the outer-training data selects the fusion penalty γ in Equation 1 by validation root mean squared error (RMSE). This simple tuning protocol is deliberate: the goal of the experiments is to compare solver representations under a common protocol, not to optimize hyperparameter-search efficiency. It was sufficient to select nontrivial fused models that outperform the featureless baseline in the reported benchmark panels. The final model is then refit on the full outer-training fold using the selected λ and γ and evaluated on the held-out outer fold. The featureless baseline predicts the training-fold mean for the corresponding response. Runtime refers to model fitting only, and memory is reported as the peak resident set size of the fitting process.

5.3 Graph Construction and Hyperparameters

The real-data benchmark constructs a data-driven group graph from training data, following the use of graph-based relationships and k-nearest-neighbor fusion graphs in prior work (Hallac et al., 2015; Padilla et al., 2020). Group centroids are computed on the outer-training fold, a k-nearest-neighbor graph is formed from centroid distances, inverse-distance weights are used, and connectivity is enforced by adding nearest inter-component links. The manifest records $k = 5$ nearest neighbors, inverse-distance weights, and MST-style connectivity augmentation. This follows the general idea that graph structure should encode similarity between groups rather than defaulting to a complete graph.

Each dataset is evaluated with five outer folds. The regularization parameter λ is warm-started from `cv.glmnet`; the fusion parameter γ is selected from the grid

$$10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}.$$

The L1 optimization tolerance is 10^{-5} with a maximum of 1200 iterations in the HPC manifest. Runs use intercepts for L1 models, no feature scaling inside the L1 solver, standardized inputs for benchmark construction, an edge-block size of 256 for the operator implementation, and the adaptive operator fallback threshold recorded in the manifest. The L2 panel uses the same outer-fold and graph protocol with the L2 solvers.

5.4 Compared Methods

For L1 fusion, the benchmark compares the legacy full construction (`old_l1`), the exact active-edge operator solver (`operator`), the chain approximation (`chain_specialized`), and a featureless baseline. For L2 fusion, the benchmark compares the legacy augmented design (`old_l2`), the active-edge augmented design (`new_l2`), and the featureless baseline.

The exact comparisons are intentionally conservative. The operator and active-edge L2 methods are compared against legacy implementations using the same graph, outer folds, selected hyperparameters, and prediction pipeline. Thus the reported differences isolate representation and construction costs, not changes in the statistical model or optimizer family. Comparisons against ADMM, split-Bregman, or other edge-splitting optimizers answer a broader question about optimizer choice; the present experiments instead test whether the sparse graph can be preserved while keeping the L1 solver within the smoothed proximal-gradient family and the L2 solver within the augmented-design plus `glmnet` family. The chain-specialized L1 method is reported separately because it changes the graph and is therefore an approximation rather than an exact replacement.

6 Results

Table 3 collects the main empirical results before the detailed panels. The structured synthetic example tests when graph fusion is statistically useful relative to unfused `glmnet` baselines, while the L1 and L2 benchmark panels test whether sparse-graph representations preserve predictive accuracy while reducing runtime and memory.

Table 3: Headline empirical findings. The changepoint experiment compares fused regression with unfused `glmnet` baselines. Solver comparisons report accuracy differences relative to the corresponding full-construction baseline, with runtime and memory summarized by medians across the completed benchmark panels.

Experiment	Comparison	Accuracy result	Efficiency or structure result
Changepoint structure	fused lasso vs. lasso(all) and lasso(same)	lowest RMSE and coefficient error	recovers exactly two large jumps at the true changepoints
Exact L1 representation	active-edge operator vs. full-edge L1	maximum mean-RMSE difference below 10^{-5} percent	2.26x median speedup and 10.7% median peak-memory reduction
Approximate L1 chain	chain-specialized solver vs. full-edge L1	2.02% median RMSE increase	3.26x median speedup and 4.6% median peak-memory reduction
Exact L2 representation	active-edge augmented design vs. full-pairwise L2	-0.025% median RMSE difference	14.9x median speedup and 19.9% median peak-memory reduction

6.1 Changepoint Structure Relative to Lasso

Before comparing solver representations, we first show a setting where fusion is statistically useful relative to unfused lasso baselines. The purpose of this experiment is not to measure scalability, but to make clear why a graph-fused regression model is useful in the first place.

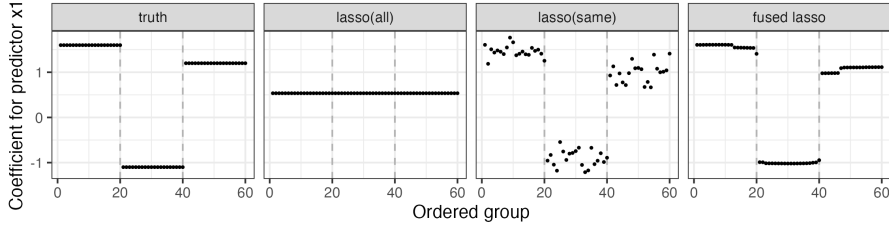


Figure 3: Controlled changepoint example comparing fused regression with unfused lasso baselines. Points show the coefficient for the first predictor across all 60 ordered groups, and dashed vertical lines mark the true changepoints. `lasso(all)` underfits group heterogeneity, `lasso(same)` overfits noisy group-level paths, and the fused lasso model recovers the piecewise-constant coefficient structure. The lasso baselines are fit with `glmnet`.

The data-generating process has ordered groups whose regression coefficients are neither globally identical nor unrelated across groups. The intended structure is piecewise constant across three contiguous blocks: groups 1–20, groups 21–40, and groups 41–60 each have stable coefficients, with changes only at the two boundaries between these blocks.

We generated an ordered-group regression problem with $k = 60$ groups and $p = 20$ predictors. The true coefficient paths are piecewise constant: groups 1–20 share one coefficient block, groups 21–40 share a second block, and groups 41–60 share a third block. Thus the two true changepoints occur after groups 20 and 40. Each group has 25 training observations for fitting, 10 validation observations for tuning, and 80 test observations. The fusion graph is the chain over the ordered groups, so the fused model penalizes differences only between neighboring groups.

We compare three estimators. The `lasso(all)` baseline fits one shared coefficient vector using all groups together, so it is stable but cannot represent group-level changepoints. The `lasso(same)` baseline expands the design with group-feature interactions and fits separate group coefficients without any fusion penalty; it can represent heterogeneity, but it does not borrow strength across neighboring groups. The fused lasso model also estimates group-level coefficient vectors, but adds an L1 fusion penalty on adjacent groups in the chain. Both lasso baselines are fit with `glmnet`.

Figure 3 shows the fitted path for the first predictor across the 60 ordered groups. The `lasso(all)` path is nearly flat because it must use the same coefficient vector for all groups. The `lasso(same)` path varies by group, but without fusion it produces a noisy path with many spurious jumps. The fused lasso path is piecewise constant and changes at the two true changepoints. Table 4 reports the same conclusion numerically: the fused lasso has the lowest test RMSE, the lowest coefficient error, and exactly two large estimated jumps, both at the true changepoints.

Table 4: Synthetic structured-heterogeneity comparisons against unfused lasso baselines fit with `glmnet`. The relative coefficient error is Frobenius error divided by the true coefficient Frobenius norm. CP distance is mean distance from true changepoints to the nearest estimated large jump.

Example	Method	Test RMSE	Coef. rel. error	Structure metric
Changepoint	lasso(all)	1.957	0.848	CP dist. 28; 0 jumps
Changepoint	lasso(same)	0.915	0.279	CP dist. 0; 59 jumps
Changepoint	fused lasso	0.708	0.079	CP dist. 0; 2 jumps

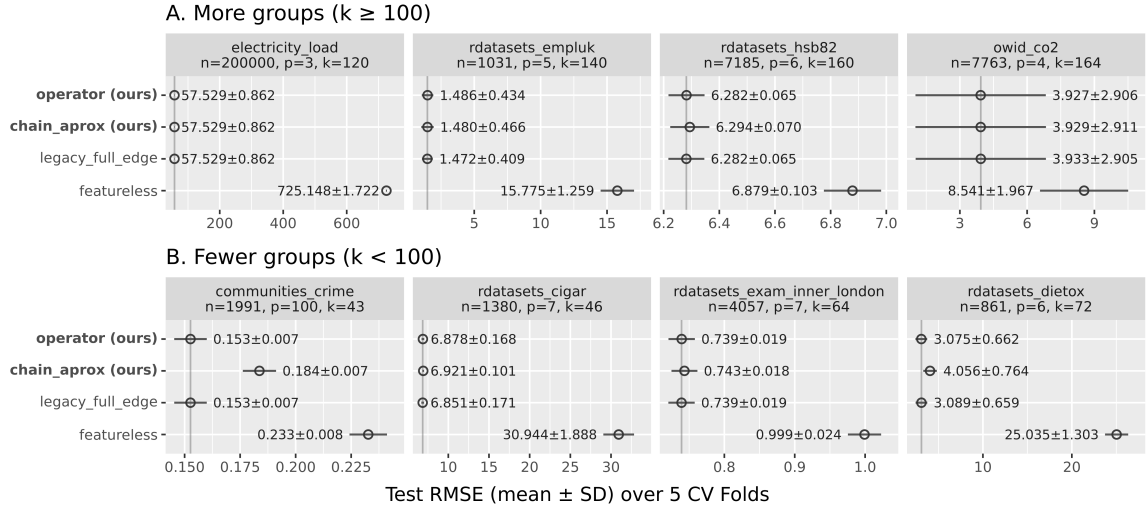


Figure 4: L1 prediction accuracy relative to the legacy L1 solver on the main benchmark panel. The exact operator solver preserves the legacy RMSE, while the chain-specialized approximation trades graph fidelity for speed.

6.2 L1 Accuracy and Efficiency

Figure 4 compares L1 test RMSE against the legacy L1 baseline. The exact operator method matches the legacy prediction error to numerical precision across the completed benchmark summary: the maximum absolute percent difference in mean RMSE is below 10^{-5} percent. This is expected because the operator form is an exact representation of the same active-edge objective.

The chain-specialized method is faster but approximate. Its median RMSE increase relative to the legacy method is 2.02% in the completed L1 summary, with larger losses on some datasets. This behavior is consistent with the method’s design: replacing a graph by a chain changes the fusion prior.

Figure 5 summarizes L1 fit time across group-count ranges. In the completed L1 summary, the exact operator method has a median speedup of 2.26x over the legacy implementation, with speedups ranging from 0.68x to 7.65x. The one slowdown occurs on a small

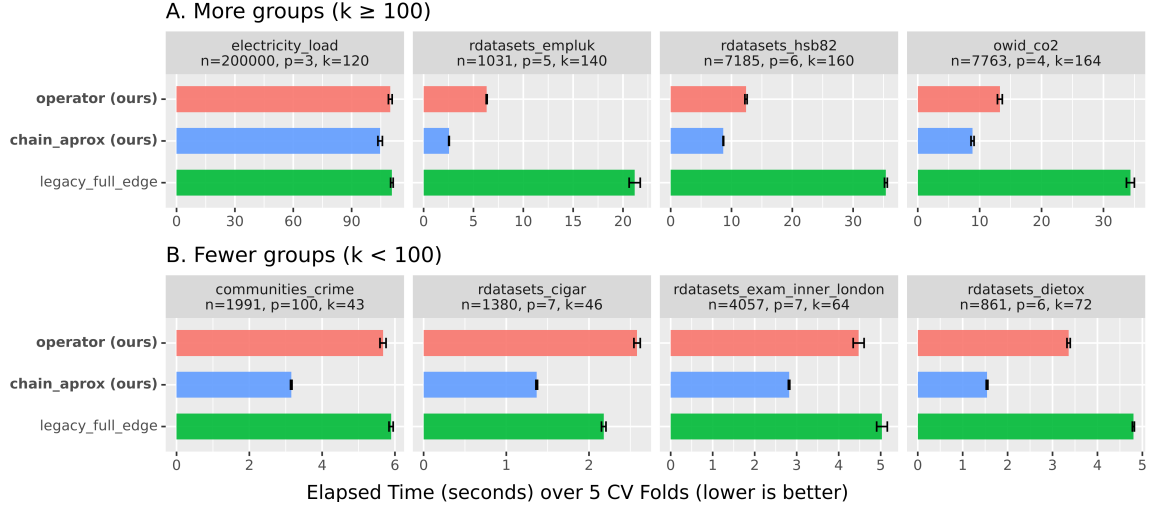


Figure 5: L1 fit time by solver and group-count range. Active-edge methods reduce the cost of graph fusion when the graph is sparse.

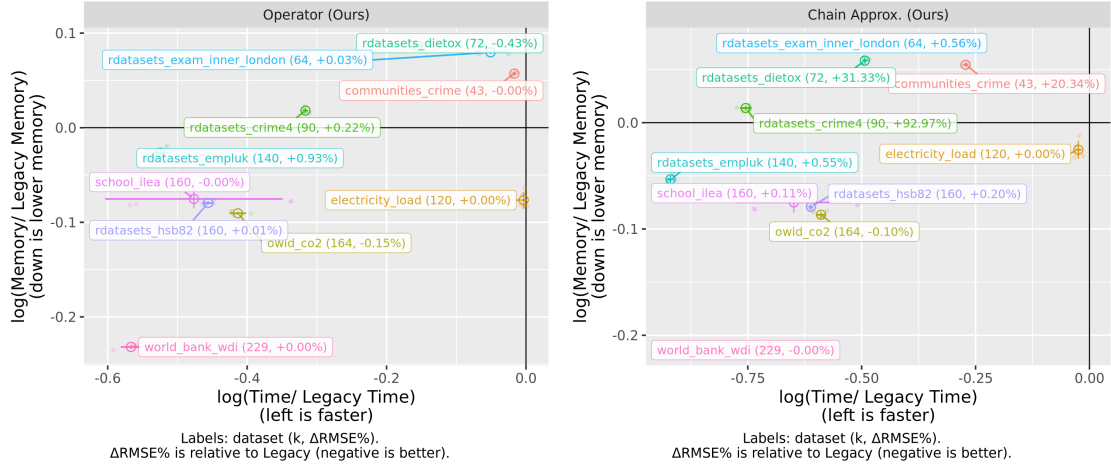


Figure 6: L1 efficiency relative to the legacy solver. Left: exact operator. Right: chain-specialized approximation.

benchmark where overhead dominates the savings from active-edge representation. Median peak memory decreases by 10.7%.

The chain-specialized approximation has a median runtime speedup of 3.26x. Its runtime gains are larger because it always uses only $k - 1$ fusion edges. Its median peak-memory reduction is 4.6%, but memory gains vary by dataset because the implementation still carries fixed R and matrix overheads.

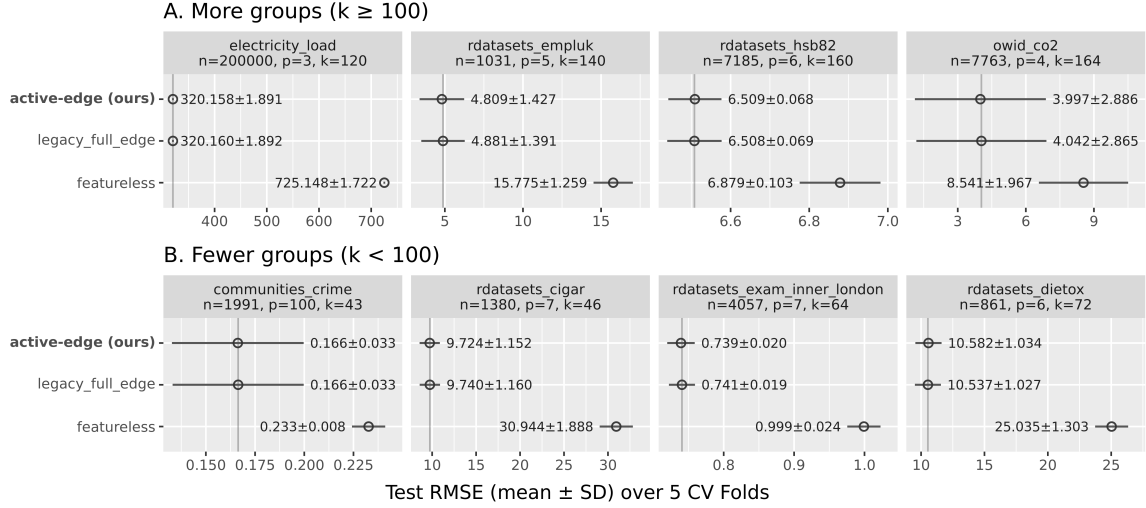


Figure 7: L2 prediction accuracy relative to the legacy L2 augmented-design solver across the benchmark panel.

6.3 L2 Accuracy and Efficiency

The L2 active-edge builder is designed to preserve the legacy augmented objective while avoiding all-pair construction. Across the 19-dataset L2 summary, **new_l2** has median percent RMSE difference of -0.025% relative to **old_l2**. The largest observed RMSE increase is 0.134%, while several datasets show small decreases in RMSE from the active-edge construction and numerical path. The median fit-time speedup is 14.9x across all 19 datasets and 105.3x on the 11 datasets with at least 50 groups. Median peak fit-worker memory decreases by 19.9% across the full L2 panel.

6.4 Synthetic Scaling Experiments

The repository also includes synthetic scaling experiments generated by the **atime** benchmark scripts. These experiments isolate solver scaling under controlled changes to the number of groups and graph structure. They support the same qualitative conclusion as the real-data benchmarks: avoiding all-pair graph construction matters most when k grows and the graph remains sparse.

7 Discussion

The main lesson is representational: graph-aware statistical models should carry graph sparsity into the optimizer. When the graph has m edges, an implementation that constructs $O(k^2)$ pairwise constraints can lose the computational advantage of modeling with a sparse graph. This is a general issue for edge-penalized regression, not only for **fuserplus**. Any method whose penalty decomposes over graph edges can pay an unnecessary quadratic cost if the graph is represented by a complete set of candidate pairs. The L1 operator and L2

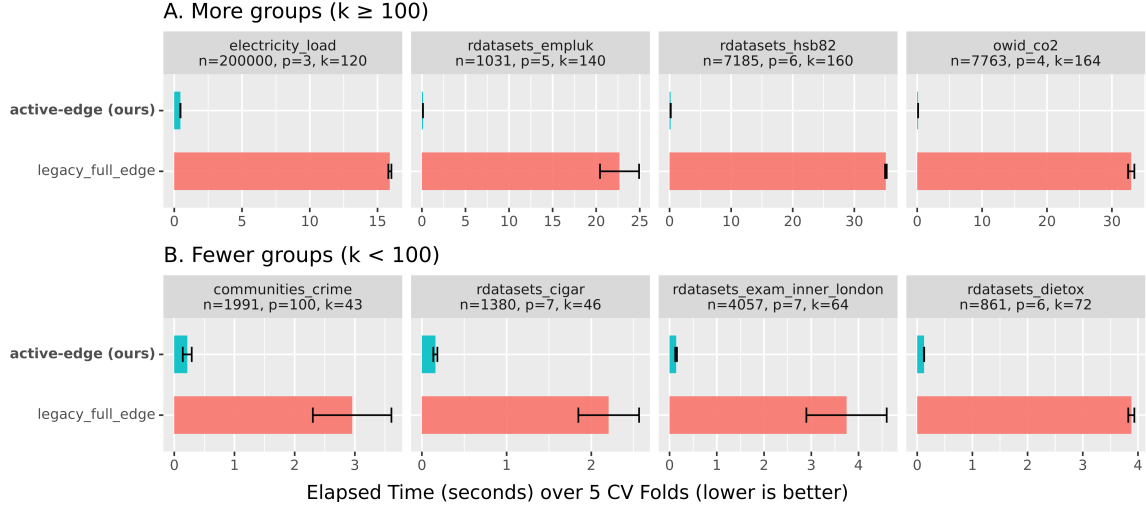


Figure 8: L2 fit time by solver and group-count range. The active-edge builder reduces augmented-design construction when the graph is sparse.

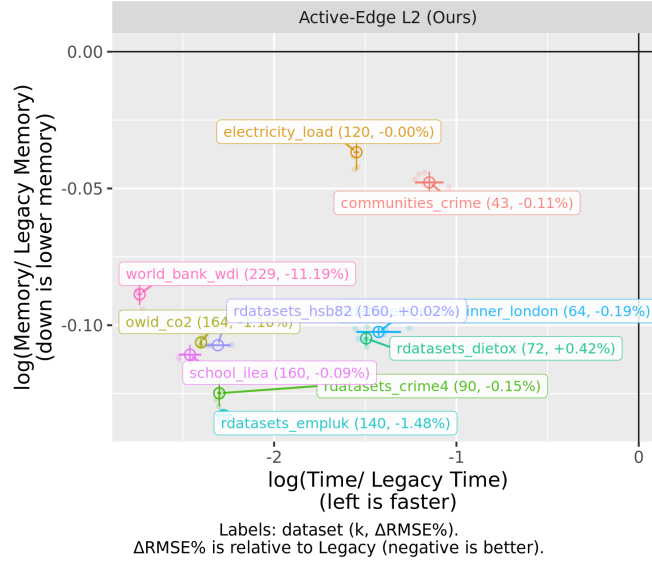


Figure 9: L2 efficiency of active-edge construction relative to the legacy L2 construction.

active-edge builders recover the intended $O(pm)$ fusion representation while preserving the original objective in the exact cases.

The exact L1 operator is the safest default among the L1 variants. It preserves the same fitted objective as the legacy active-edge formulation and gives substantial median runtime and memory improvements on the completed benchmark summary. The chain-specialized solver is more aggressive. It is best viewed as an approximation for exploratory or large-scale settings where a small loss in predictive accuracy is acceptable.

The solver representation aligns the computational graph with the modeling graph. Edge-explicit L1 fusion updates, blocked edge processing, DFS/MST chain approximation, and active-edge L2 matrix construction are instances of that representation. Proposition 1 gives the corresponding smoothed-objective convergence guarantee for the L1 APG updates, while Propositions 2 and 3 identify the exact cases where the new representations preserve the legacy objective. Proposition 4 makes the representation-dependent graph cost explicit. The chain-specialized method is separated from those exact methods because replacing a graph by a chain changes the fusion geometry; Proposition 5 states the precise chain objective being optimized and the omitted-edge penalty that accounts for the approximation.

The empirical evidence is strongest for the exact L1 operator and active-edge L2 construction. The chain variant is included because it is a useful solver mode in the package, but it involves an additional graph-approximation choice. For that reason, the recommended default for sparse L1 fusion is the exact active-edge operator, with chain-specialized fitting reserved for exploratory or memory-constrained settings.

7.1 When Sparse-Graph Representations Help

The sparse-graph representation is most useful when the modeling graph has far fewer active edges than the complete graph. The estimators considered in this paper allow arbitrary fusion graphs, including complete graphs. If the intended graph is complete, then $m = \binom{k}{2}$ and active-edge construction remains correct but does not reduce the asymptotic graph-representation cost. The computational gains therefore come from settings where the graph encodes a structured notion of similarity, such as temporal adjacency, spatial neighborhoods, nearest-neighbor group similarity, or task relationships. For complete or nearly complete graphs, the only L1 variant in this paper that can still reduce the fusion edge budget substantially is the chain-specialized approximation. It does so by replacing the original graph with a connected chain, so the gain comes from changing the optimization target rather than from an exact sparse representation of the same graph. The empirical results therefore treat the chain method separately: it can be much faster, but it may sacrifice some prediction accuracy because it omits many original fusion edges.

This distinction is separate from the statistical question of which graph is best for prediction. A complete graph may be appropriate when all group pairs should be eligible for fusion. In contrast, sparse graphs are appropriate when domain knowledge or training-fold similarity suggests that only a limited set of group relationships should be penalized directly. The contribution of the active-edge solvers is to make this sparse modeling choice visible to the optimizer: when the graph is sparse, computation and memory scale with m rather than with all $\binom{k}{2}$ possible group pairs.

7.2 Scope of the Optimization Comparisons

The experiments intentionally compare the new representations against the legacy grouped fused-regression solvers rather than against a broad collection of alternative optimizer families. This choice is central to the empirical design. The L1 legacy solver and the exact active-edge operator both use the same smoothed proximal-gradient framework; the L2 legacy and active-edge solvers both use the same augmented-design reduction followed by `glmnet`. The comparisons therefore hold fixed the statistical objective, graph, folds, hy-

perparameter selection, prediction pipeline, and optimizer family. The primary changing factor is the representation of the graph-fusion term: dense or all-pair construction versus active-edge construction. This isolation is what makes the empirical results interpretable as evidence for the sparse-graph representation principle.

Alternative optimizers, including ADMM and split-Bregman methods, are important for fused-lasso problems and can be very effective in other settings. They answer a different question. An ADMM formulation would typically introduce auxiliary variables for edge differences, enforce $z_{uv} = \beta_u - \beta_v$ with augmented-Lagrangian updates, alternate between coefficient, edge-variable, and dual updates, and require choices of penalty parameters and stopping rules. Such a comparison would mix representation effects with optimizer effects: linear-system solves or preconditioners, penalty-parameter tuning, convergence tolerances, warm starts, sparse-matrix backends, and implementation language could all influence run-time and memory. Those factors are scientifically interesting, but they would not isolate the question studied here.

This distinction also clarifies the role of the legacy fused-lasso baseline. It is not included because it is the strongest possible optimizer for every graph-fused problem. It is included because it is the direct predecessor for the grouped fused-regression objective and software interface considered in this paper. Beating that baseline while preserving the same optimizer family shows that the gain comes from avoiding zero-weight all-pair graph construction rather than from replacing the optimization architecture. The same active-edge principle could be incorporated into ADMM or other edge-splitting methods, where the number of auxiliary edge variables and dual updates would also scale with the represented edge set. Developing and tuning such solvers is a separate algorithmic comparison and is left for future work.

There are three important limitations. First, the guarantees are algorithmic: they cover representation equivalence, smoothing error, convergence for the smoothed L1 objective, and graph-construction complexity. The statistical properties are those of the underlying fused-lasso and graph-guided regression objectives, while the contribution here is to preserve those objectives with a sparser computational representation. Second, the largest gains occur when the user-supplied or data-derived graph is sparse. If the modeling graph is close to complete, the active-edge representation has little edge-count advantage. Third, the implementation used in the experiments is written in R, so small problems can be dominated by fixed interpreter, allocation, and `glmnet` overheads. These limits explain the occasional small-benchmark slowdown and motivate the recommendation to use the exact operator as the default sparse-graph path rather than as a universal speed guarantee.

8 Reproducibility

The algorithms are implemented in the `fuserplus` R package, version 0.1.0, released under the MIT license. The project repository and issue tracker are available at <https://github.com/EngineerDanny/fuserplus>. The experiments in this paper were evaluated from repository commit prefix `99c69ae`. The benchmark environment used R version 4.5.2 (2025-10-31). The package includes documentation, a vignette, and `testthat` coverage for the solver routines used in the experiments.

The real-data benchmark summaries used in the figures are `build/hpc/final_test_summary.csv` for the L1 panel and `build/hpc/l2_knn/final_test_summary.csv` for the L2 panel. The structured synthetic example is generated by the changepoint benchmark script under `scripts`. The manifest-based benchmark scripts are stored under `scripts/hpc`, and the paper source is `publication/paper1/fuserplus_jmlr.tex`.

9 Conclusion

Sparse-graph construction can materially improve grouped fused regression solvers without changing the statistical model. The exact L1 operator and active-edge L2 augmented-design builder preserve the legacy objectives while reducing avoidable all-pairs graph costs. The chain-specialized L1 approximation provides an additional high-efficiency mode when the user is willing to trade graph fidelity for speed. These results support a broader principle for graph-penalized regression: when the penalty is defined on a sparse graph, the optimizer should preserve the edge-level representation rather than encoding sparsity inside a dense all-pairs construction.

Appendix A. Supplementary Dataset Inventory

Table 5 lists the complete union of datasets that appear in the final L1 and L2 result summaries used by this paper. The `SubsetType` column reports the semantic unit defining the subgroups in the regression problem after preprocessing in this repository.

Table 5: Public grouped-regression datasets used in the reported real-data benchmark panels. The columns report total observations n , feature count p , group count k , and semantic subgroup type after preprocessing in this repository.

Dataset	n	p	k	SubsetType	Source
beijing_air_quality	180,000	14	12	stations	UCI Beijing air quality (Chen, 2017)
communities_crime	1,991	100	43	states	UCI Communities and Crime (Redmond, 2002)
electricity_load	200,000	3	120	meters	UCI Electricity Load (Trindade, 2015)
nyc_tlc_yellow_2022	200,000	9	82	pickup zones	NYC TLC yellow taxi records (New York City Taxi and Limousine Commission, 2022)
owid_co2	7,763	4	164	countries	Our World in Data CO2 (Ritchie et al., 2023)
radon_minnesota	916	5	82	counties	PyMC Minnesota radon example (PyMC Development Team, 2026)
rdatasets_cigar	1,380	7	46	states	Rdatasets, <code>plm::Cigar</code> (Arel-Bundock, 2026)
rdatasets_crime4	630	57	90	counties	Panel crime data (Cornwell and Trumbull, 1994)
rdatasets_dietox	861	6	72	pigs	Dietox pig-growth data (Lauridsen et al., 1999)
rdatasets_empluk	1,031	5	140	firms	Employment panel data (Arellano and Bond, 1991)
rdatasets_exam_inner_london	4,057	7	64	schools	Inner London exam data (Goldstein et al., 1993)
rdatasets_fatalities	336	32	48	states	Rdatasets, <code>AER::Fatalities</code> (Arel-Bundock, 2026)
rdatasets_grunfeld	200	3	10	firms	Rdatasets, <code>plm::Grunfeld</code> (Arel-Bundock, 2026)
rdatasets_guns	1,173	11	51	states	Rdatasets, <code>AER::Guns</code> (Arel-Bundock, 2026)
rdatasets_hsb82	7,185	6	160	schools	High-school multilevel data (Raudenbush and Bryk, 2002)
rdatasets_ratpupweight	322	3	27	litters	Rdatasets, <code>nlme::RatPupWeight</code> (Arel-Bundock, 2026)
sarcos_openml	44,484	21	10	response bins	OpenML SARCOS 43873 (OpenML, 2026)
school_ilea	7,185	10	160	schools	<code>nlme</code> school data (Pinheiro and Bates, 2000)
world_bank_wdi	10,830	4	229	economies	World Development Indicators (World Bank, 2026)

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